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SSRS Operational Reporting - Conceptual Implementation Document

Objective:

To design operational reports using SQL Server Reporting Services (SSRS) that display **ticket volume, resolution time, and SLA compliance**, enabling customer support managers to monitor day-to-day service operations effectively.

The conceptual implementation ensures:

- Parameterized reports
- Clean formatting and grouping
- Accurate mapping to SQL data sources

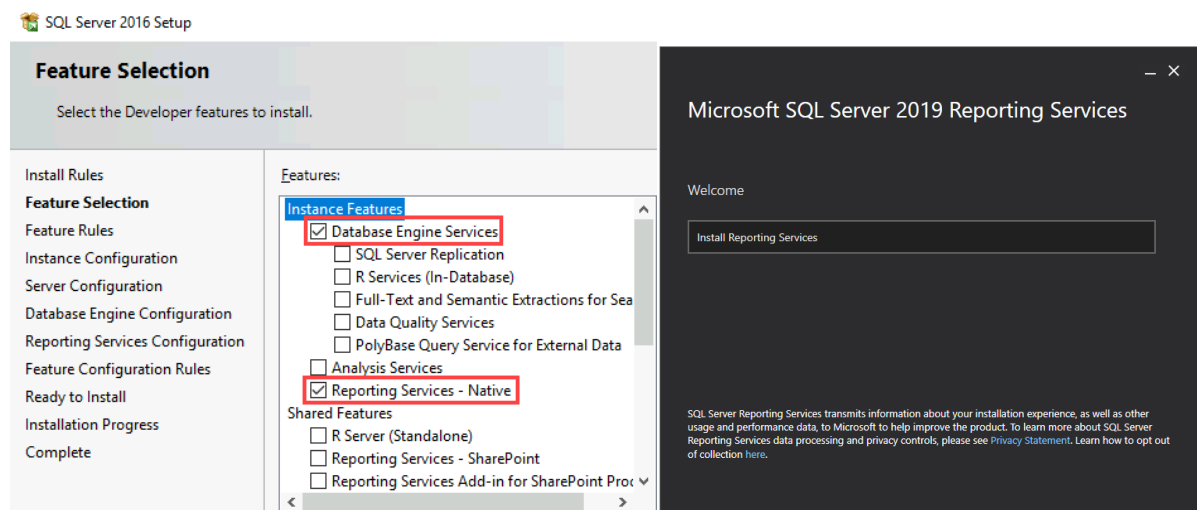
Step 1 — Install SQL Server Reporting Services (SSRS)

SSRS is installed as a standalone reporting platform used to design, deploy, and manage reports.

Installation includes:

- Report Server engine
- Web portal
- Management tools

This step prepares the environment for report creation.



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Step 2 — Connect to Report Server

After installation, the Reporting Services Configuration Manager is used to connect to the report server instance.

Key actions:

- Specify server name
- Select report server instance
- Establish administrative connection

This enables configuration and deployment capabilities.

The screenshot shows the 'Web Service URL' configuration window in the Reporting Services Configuration Manager. The left-hand navigation pane lists several configuration areas: 'Connect', 'Server Account', 'Web Service URL' (which is highlighted with an orange border), 'Database', 'Report Manager URL', 'E-mail Settings', 'Execution Account', 'Encryption Keys', and 'Scale-out Deployment'. The main pane on the right is titled 'Web Service URL' and contains the following configuration options:

- Report Server Web Service Virtual Directory:** A text box containing 'ReportServer'.
- Report Server Web Service Site identification:**
 - Virtual Directory:** A dropdown menu set to 'All Assigned (Recommended)'.
 - TCP Port:** A text box containing '80'.
 - SSL Certificate:** A dropdown menu set to '(Not Selected)', which is highlighted with an orange border.
 - SSL Port:** An empty text box.
- Report Server Web Service URLs:**
 - URLs:** A text box containing the URL http://USER1_LENOVO:80/ReportServer.

Step 3 — Configure Report Server Database

SSRS requires a backend database to store:

- Report definitions
- Metadata
- Security information
- Execution logs

A new ReportServer database is created on SQL Server.

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The image shows two side-by-side windows from the Microsoft Reporting Services configuration tools.

The left window is the "Report Server Database Configuration Wizard" with the "Change Database" step selected. It prompts the user to "Choose whether to create or configure a report server database." The "Database Server" tab is active, showing fields for "Server Name" (M10A0104), "Authentication Type" (Current User - Integrated Security), "Username" (ADOS\viktor.szentes), and "Password". A "Test Connection" button is at the bottom right.

The right window is the "Web Service URL" configuration page. It has a left-hand "Connect" pane with options like "Server Account", "Web Service URL" (highlighted with an orange box), "Database", "Report Manager URL", "E-mail Settings", "Execution Account", "Encryption Keys", and "Scale-out Deployment". The main area contains the following settings:

- Report Server Web Service Virtual Directory:** Virtual Directory: ReportServer
- Report Server Web Service Site identification:** Virtual Directory: All Assigned (Recommended), TCP Port: 80, SSL Certificate: (Not Selected) (highlighted with an orange box), SSL Port: (empty)
- Report Server Web Service URLs:** URLs: http://USER1_LENOVO/80/ReportServer

Step 4 — Configure Web Service URL

The Web Service URL exposes the report server for programmatic access and deployment.

Typical configuration:

- HTTP endpoint creation
- Port configuration
- Service binding

This URL is used by development tools to publish reports.

This screenshot shows the "Web Service URL" configuration page with the "Web Service URL" option selected in the left-hand "Connect" pane. The main configuration area includes:

- Report Server Web Service Virtual Directory:** Virtual Directory: ReportServer
- Report Server Web Service Site identification:** IP Address: (dropdown), TCP Port: (empty), SSL Certificate: (dropdown), SSL Port: (empty), and an "Advanced..." button.
- Report Server Web Service URLs:** URLs: <http://localhost:80/ReportServer> and <https://<server name>:443/ReportServer> (the second URL is highlighted with a red box).

A "Results" section is visible at the bottom of the window.

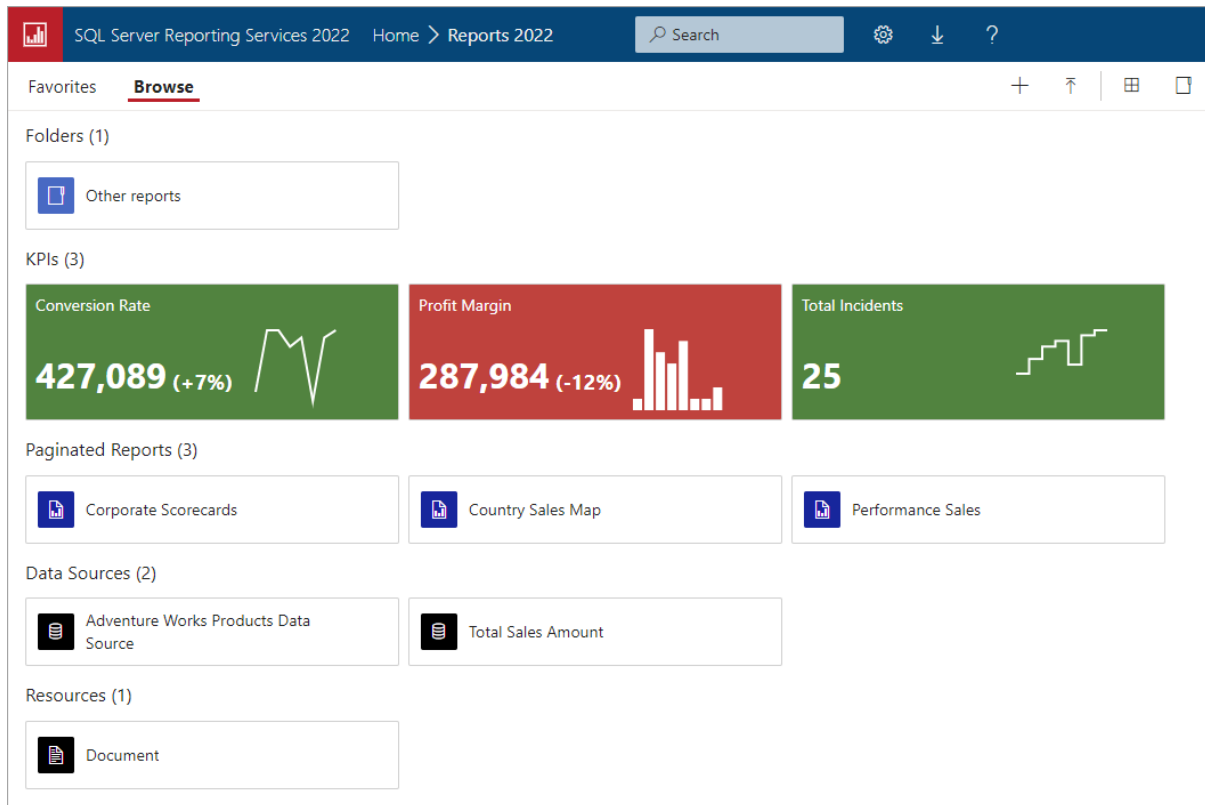
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Step 5 — Configure Web Portal URL

The Web Portal provides a user interface for viewing and managing reports.

Support managers access reports through this portal to monitor operations.



Step 6 - Create Report Server Project in Visual Studio

A Report Server Project is created using SQL Server Data Tools.

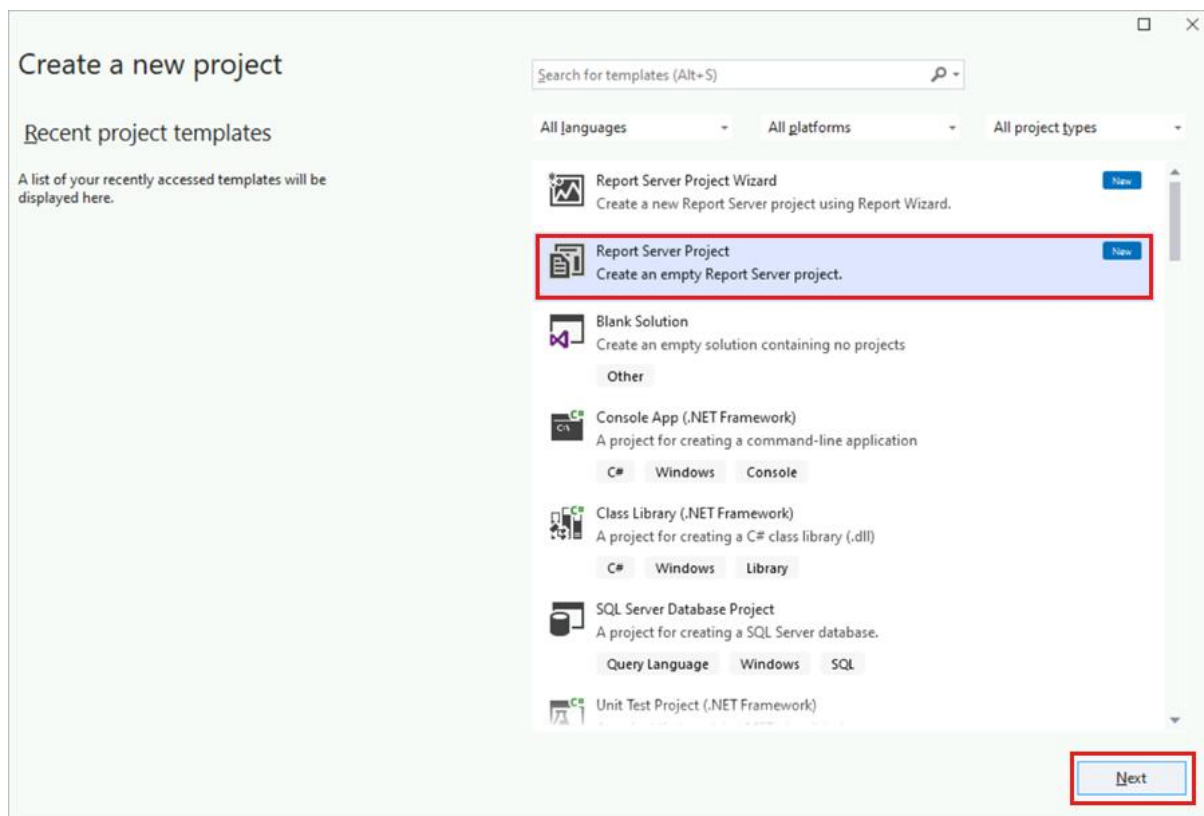
Project structure includes:

- Shared Data Sources
- Shared Datasets
- Reports

This enables organized development and deployment.

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Step 7 — Configure Shared Data Source

The report connects to the operational database (Customer Engagement System).

Configuration includes:

- SQL Server instance
- Authentication method
- Target database

Data Source Mapping

Database: **CustomerEngagementDB**

Tables Used:

- Tickets
- Customers
- Agents
- TicketCategories

This ensures accurate mapping to SQL data sources (acceptance criterion).

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The screenshot shows the 'Data Source Properties' dialog box with the 'General' tab selected. The 'Name' field is 'DataSource1'. The 'Embedded connection' radio button is selected, and the 'Type' is 'Microsoft SQL Server'. The 'Connection string' field is empty, with a 'Build...' button and a formula icon next to it. The 'Use shared data source reference' radio button is unselected, and the 'Use single transaction when processing the queries' checkbox is also unselected. The 'OK' button is highlighted with a blue border.

Data Source Properties

General

Credentials

Change name, type, and connection options.

Name:
DataSource1

☒ Embedded connection:

Type:
Microsoft SQL Server

Connection string:
Click here to type or paste a connection string

Build...

☐ Use shared data source reference

Edit... New...

☐ Use single transaction when processing the queries

Help OK Cancel

Step 8 — Design Dataset Queries

Datasets retrieve data required for reports.

◆ Ticket Volume Query

```
SELECT Status, COUNT(*) AS TicketCount
FROM Tickets
GROUP BY Status;
```

Purpose: Monitor workload distribution.

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◆ Resolution Time Query

```
SELECT
AVG(DATEDIFF(hour, CreatedAt, ResolvedAt)) AS AvgResolutionHours
FROM Tickets
WHERE ResolvedAt IS NOT NULL;
```

◆ SLA Compliance Query

Assumption: SLA = 48 hours

```
SELECT
COUNT(*) AS SLA_Compliant_Tickets
FROM Tickets
WHERE DATEDIFF(hour, CreatedAt, ResolvedAt) <= 48;
```

Purpose: Monitor adherence to service agreements.

Step 9 — Create Parameterized Reports

To meet acceptance criteria, reports include parameters such as:

- Date range
- Ticket category
- Agent
- Status

Example parameterized query:

```
SELECT *
FROM Tickets
WHERE CreatedAt BETWEEN @StartDate AND @EndDate;
```

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Benefits:

- Dynamic filtering
- Interactive reporting
- Customized operational insights

Step 10 — Apply Formatting and Grouping

Reports are formatted for managerial readability:

Formatting Features

- Clear headers
- Tabular layouts
- KPI indicators
- Alignment and spacing

Grouping Examples

- Tickets grouped by status
- Performance grouped by agent
- Trends grouped by date

This satisfies the “clear formatting and grouping” criterion.

Step 11 — Add Visual Enhancements

SSRS supports visual components such as:

- Charts (bar, pie, line)
- KPI indicators
- Conditional formatting

Example:

- Red highlight for SLA breaches
- Green for compliant tickets

Step 12 — Preview and Validate Report

Reports are previewed to verify:

- Data accuracy
- Parameter functionality
- Layout correctness
- Performance

Step 13 — Deploy Report to Server

Deployment publishes reports to the SSRS portal for user access.

Support managers can then monitor operations in real time.

Step 14 — Operational Usage

Customer support managers use the reports to:

- Track ticket backlog
- Evaluate resolution efficiency
- Monitor SLA risks
- Identify bottlenecks

This fulfills the user story requirement.

Acceptance Criteria Coverage

- ✓ Parameterized SSRS reports — implemented via dynamic filters
- ✓ Clear formatting and grouping — structured layout and visual design
- ✓ Accurate mapping to SQL data sources — direct connection to operational DB

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Conclusion

The conceptual SSRS implementation provides a scalable reporting framework for monitoring service operations. By integrating ticket volume, resolution metrics, and SLA compliance into structured reports, the system enables data-driven decision-making and continuous service improvement.